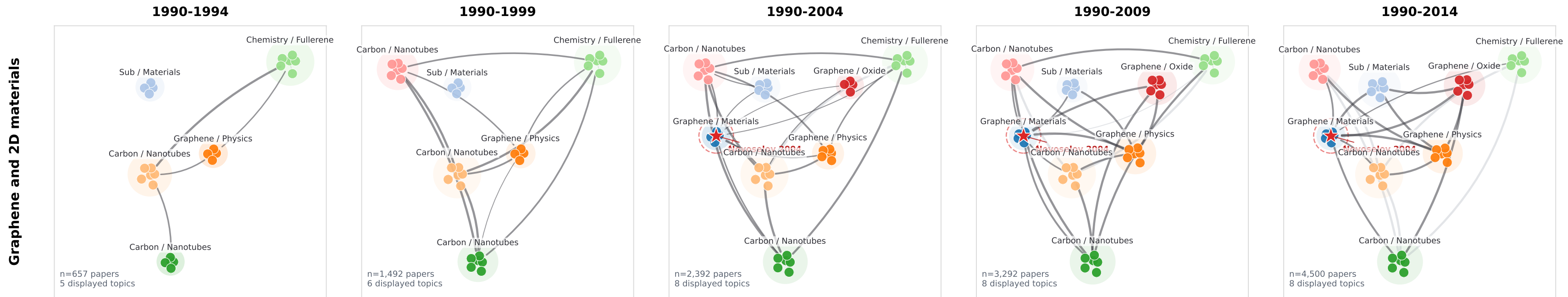


Fig. 1 | Citation knowledge-graph perturbation after a landmark innovation

Example domain: Graphene and 2D materials. Real OpenAlex data; publication-style display uses data-derived topic snapshots.



a Cumulative citation knowledge graph snapshots



Carbon nanomaterial priors

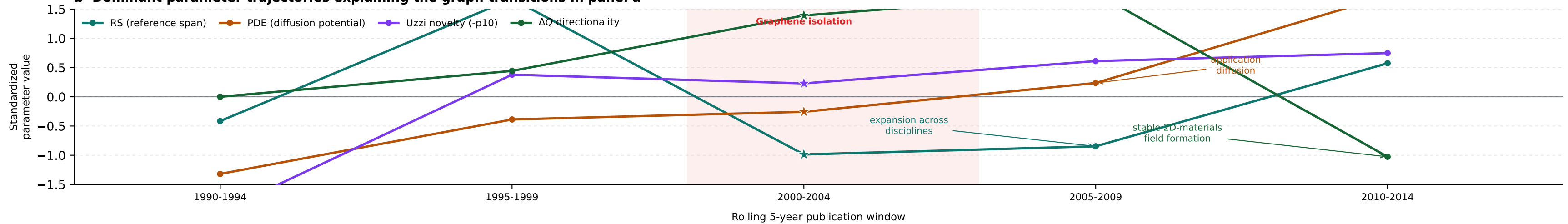
Graphite and nanotube consolidation

Graphene isolation shock

2D-materials expansion

Device and heterostructure hubs

b Dominant parameter trajectories explaining the graph transitions in panel a



Expansion

RS up + PDE up

Physics, materials and devices connect

Novelty

Uzzi up

Atypical recombination of carbon physics

Field formation

DeltaQ later > 0

Stable 2D-materials subfield emerges